

KRISTIANSAND KJEVIK, Norway

WMO: 014520

Lat: 58.204N

Lon: 8.085E

Elev: 57

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.2	12.5	2.2	6.2	11.5	6.9	7.9	17.2	27.1	41.2	24.8	39.3	6.8	50	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	76.9	63.0	73.8	61.5	71.2	60.4	65.5	72.6	63.8	70.5	62.3	68.3	9.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	86.0	67.7	61.2	80.8	66.2	59.5	76.1	64.9	30.3	72.7	29.1	70.7	28.0	68.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.9	17.8	3.5	80.8	6.2	3.3	-1.0	83.1	-4.6	85.1	-8.1	86.9	-12.6	89.4		
(5)			2.9	67.5	6.1	1.9	-1.5	68.8	-5.0	70.0	-8.5	71.0	-12.9	72.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	31.7	32.1	35.9	42.7	50.6	57.3	61.9	60.7	54.7	46.4	39.0	33.1	(6)
	DBStd	12.49	8.24	7.77	6.52	5.43	5.61	4.38	4.15	4.15	4.35	5.54	6.54	8.94	(7)
	HDD50	2797	566	500	438	226	59	2	0	0	11	137	333	525	(8)
	HDD65	7123	1031	920	903	668	446	235	116	146	311	576	781	990	(9)
	CDD50	1186	0	0	1	8	79	221	369	331	150	25	2	0	(10)
	CDD65	35	0	0	0	0	1	4	19	11	0	0	0	0	(11)
	CDH74	243	0	0	0	0	4	33	139	66	1	0	0	0	(12)
	CDH80	14	0	0	0	0	0	4	7	3	0	0	0	0	(13)
(14) Wind	WSAvg	8.3	8.9	8.9	8.7	8.4	8.5	8.3	7.9	7.7	8.1	8.0	8.4	8.3	(14)
(15) Precipitation	PrecAvg	54.4	6.0	3.9	3.1	2.8	3.0	3.3	3.3	5.0	5.4	6.5	6.5	5.4	(15)
	PrecMax	76.4	16.1	11.3	6.0	7.1	6.9	8.1	8.7	10.0	11.8	12.5	17.4	11.5	(16)
	PrecMin	30.6	1.0	0.5	0.4	0.4	0.2	0.1	0.1	0.4	1.4	2.2	1.6	1.3	(17)
	PrecStd	10.9	3.2	2.5	1.5	1.6	1.7	2.0	2.0	2.3	2.6	2.8	3.5	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	51.8	57.0	64.4	74.2	77.9	80.9	79.1	70.2	60.8	55.4	50.9	(19)
		MCWB	44.5	45.8	47.8	51.6	59.4	60.8	64.4	63.6	59.8	54.3	51.4	47.7	(20)
	2%	DB	45.8	46.6	51.9	59.8	69.7	73.4	77.9	75.5	66.5	57.9	52.8	48.6	(21)
		MCWB	42.9	42.5	43.6	49.4	56.8	59.3	63.5	63.2	58.2	53.6	49.7	46.2	(22)
	5%	DB	44.2	44.6	48.1	55.8	65.0	69.9	74.8	72.7	64.4	56.6	50.6	47.0	(23)
		MCWB	41.8	41.4	41.4	46.8	54.2	57.8	62.4	62.2	57.6	53.0	47.9	44.8	(24)
	10%	DB	42.5	42.7	45.0	52.2	61.4	67.0	71.6	69.6	62.4	55.2	48.9	45.0	(25)
		MCWB	40.3	40.1	40.1	44.5	52.2	56.5	60.7	61.1	56.8	51.8	46.5	42.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	46.8	48.4	54.3	61.9	63.8	68.1	67.5	63.6	57.2	52.6	48.4	(27)
		MCDB	47.1	50.4	54.6	61.5	71.0	72.3	75.9	73.1	66.8	58.9	54.3	49.8	(28)
	2%	WB	43.5	43.9	44.8	50.4	58.5	61.6	65.8	65.5	61.1	55.5	50.3	46.8	(29)
		MCDB	45.0	45.4	49.5	57.1	66.1	69.9	73.6	71.8	64.3	57.1	52.1	48.2	(30)
	5%	WB	42.0	42.0	42.8	48.3	55.7	59.7	64.1	63.8	59.3	54.3	48.6	45.1	(31)
		MCDB	43.4	43.6	46.3	54.1	62.9	66.9	71.5	69.9	62.5	56.1	50.2	46.6	(32)
	10%	WB	40.5	40.4	41.2	46.1	53.4	58.0	62.6	62.3	57.9	52.7	46.9	43.0	(33)
		MCDB	42.0	42.2	44.2	50.3	59.8	64.2	69.1	67.6	61.2	54.8	48.6	44.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.8	12.7	14.5	15.4	14.8	15.0	14.5	13.0	11.7	10.1	10.0	(35)
		MCDBR	11.2	12.2	17.9	20.1	21.5	20.1	21.0	19.0	15.9	12.9	11.4	10.2	(36)
	5% WB	MCWBR	10.0	9.2	11.2	11.6	11.4	10.6	10.0	9.7	9.9	9.5	10.4	9.6	(37)
		MCWBR	10.0	10.5	14.7	17.0	18.8	17.4	17.7	15.8	12.7	10.1	10.4	10.1	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.293	0.321	0.339	0.384	0.374	0.360	0.379	0.380	0.357	0.334	0.304	0.271	(40)
		taud	2.244	2.326	2.370	2.300	2.404	2.466	2.430	2.423	2.479	2.459	2.313	2.193	(41)
	Ebn at Noon	Ebn at Noon	172	226	257	261	273	279	270	260	249	220	168	141	(42)
		Edh at Noon	15	22	28	35	34	32	33	31	25	19	14	11	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	108	290	731	1230	1659	1872	1700	1305	845	395	146	75	(44)
	RadStd	RadStd	23	60	71	152	188	151	172	92	93	61	33	10	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	-216	-290	N/A	N/A

Nomenclature: See separate page